Graph Irregularity via Edge Deletions

Julien Bensmail[1], Noëmie Catherinot[2], Foivos Fioravantes[3(✉)], Clara Marcille[4], and Nacim Oijid[4,5]

[1] Université Côte d'Azur, CNRS, Inria, I3S, Côte d'Azur, France
[2] DER Informatique, Université Paris-Saclay, ENS Paris-Saclay, 4 Avenue des Sciences, 91190 Gif-sur-Yvette, France
noemie.catherinot@ens-paris-saclay.fr
[3] Department of TCS, FIT, Czech Technical University in Prague, Prague, Czech Republic
foivos.fioravantes@fit.cvut.cz
[4] Univ. Bordeaux, CNRS, Bordeaux INP, LaBRI, UMR 5800, 33400 Talence, France
nacim.oijid@ens-lyon.fr
[5] Department of Mathematics and Mathematical Statistics, Umeå University, Umeå, Sweden

Abstract. We pursue the study of edge-irregulators of graphs, which were recently introduced in [Fioravantes et al. Parametrised Distance to Local Irregularity. *IPEC*, 2024]. That is, we are interested in the parameter $\mathrm{I}_e(G)$, which, for a given graph G, denotes the smallest $k \geq 0$ such that G can be made locally irregular (*i.e.*, with no two adjacent vertices having the same degree) by deleting k edges. We exhibit notable properties of interest of the parameter I_e, in general and for particular classes of graphs, together with parameterized algorithms for several natural graph parameters.

Despite the computational hardness previously exhibited by this problem (NP-hard, W[1]-hard w.r.t. feedback vertex number, W[1]-hard w.r.t. solution size), we present two FPT algorithms, the first w.r.t. the solution size plus Δ and the second w.r.t. the vertex cover number of the input graph.

Finally, we take important steps towards better understanding the behaviour of this problem in dense graphs. This is crucial when considering some of the parameters whose behaviour is still uncharted in regards to this problem (e.g., neighbourhood diversity, distance to clique). In particular, we identify a sub-family of complete graphs for which we are able to provide the exact value of $\mathrm{I}_e(G)$. These investigations lead us to propose a conjecture that $\mathrm{I}_e(G)$ should always be at most $\frac{1}{3}m + c$, where m is the number of edges of the graph G and c is some constant. This conjecture is verified for various families of graphs, including trees.

Due to space limitations, many proofs are omitted or just sketched. We refer the reader to the full version [6]. This research was partly funded by Kempe Foundation Grant No. JCSMK24-515 (Sweden) and the International Mobility of Researchers MSCA-F-CZ-III at CTU in Prague, CZ.02.01.01/00/22_010/0008601 Programme Johannes Amos Comenius.

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026
E. Di Giacomo and D. Mondal (Eds.): WALCOM 2026, LNCS 16444, pp. 353–367, 2026.
https://doi.org/10.1007/978-981-95-7127-7_24

1 Introduction

In graph theory, the family of *regular graphs* is used to capture the most common notion of regularity, that of all vertices having the same degree. A very natural question then, is on possible definitions of graphs that would be sort of antonyms to regular graphs. This led several groups of researchers to investigate various definitions of irregularity in graphs, such as *irregular graphs* (graphs in which all degrees are pairwise distinct [7]), *highly irregular graphs* (graphs in which no vertex has two neighbours with the same degree [1]), and *locally irregular graphs* (graphs in which no two adjacent vertices have the same degree [2]). As we will only focus on the latter notion throughout this work, we encourage the reader to refer to the literature for more details on the former two.

Note that most definitions of (ir)regular graphs do not hold for all simple graphs: a given simple graph can be more or less (ir)regular w.r.t. some notion of (ir)regularity, and exhibit different behaviour for different such notions. This leads to different questions, such as how a given simple graph can be turned into a (ir)regular structure through elementary operations. See, e.g., [4] for an example of a problem where regularity is introduced in the given graph through vertex deletions. We will focus on introducing irregularity. The initiators of this line of research are Chartrand *et al.* [7], who introduced the notion of *irregularity strength*. In brief, this is a parameter that measures how easy/hard it is for a given graph to be turned into an irregular multigraph by increasing the multiplicity of its edges (with the additional objective that the maximum number of parallel edges replacing a given edge should be minimised). This concern was considered as well for the other notions of irregularity, namely in [5] for highly irregular graphs, and in [13] for locally irregular graphs. These investigations resulted in numerous related questions and problems, such as the so-called 1–2-3 Conjecture (introduced by Karoński, Łuczak, and Thomason in [13], and solved recently by Keusch in [14]) for the latter notion.

In all these considerations, a given simple graph is intended to be made irregular by essentially adding edges locally (in a parallel way, upon original edges), and there is an implicit postulate: the number of edges added this way should be minimised. This leads us to the question of whether one could consider making simple graphs somewhat irregular through other elementary operations that would limit even further the eventual number of edges. In particular, could this be achieved without adding any additional edge, by just *moving* edges? We understand this operation as removing said edge and placing a new edge between any two vertices we chose (regardless of whether there already exists an edge between the considered vertices). At this point it is also worth mentioning [3], where the authors consider a similar operation on the edges but with the goal of introducing regularity to the given graph.

Transforming a given graph into a locally irregular graph via edge moves is straightforward, since any multistar on at least 3 vertices is always locally irregular. Things become more interesting when we integrate concerns borrowed from reconfiguration problems, such as minimising the number of edge moves performed. This latter problem is way harder to comprehend. Indeed, when

edge moves can be performed at will, it is important to consider the number of edges in a largest (non-induced) locally irregular subgraph of G. This leads us to the study of the (dual) concepts and notions we will focus on in this paper, which are about the minimum number of edges one must remove from a graph to make it locally irregular.

These notions were introduced recently by Fioravantes, Melissinos, and Triommatis [11]. Following their terminology, we define an *edge-irregulator* of a graph G as a set $S \subseteq E(G)$ of edges of G such that $G - S$ is locally irregular. We also set $\mathrm{I}_e(G)$ as the smallest $k \geq 0$ such that G admits edge-irregulators of cardinality k, and say that an edge-irregulator S of G is *optimal* if $|S| = \mathrm{I}_e(G)$. The authors of [11] introduced edge-irregulators as an edge counterpart of *vertex-irregulators* (introduced in [12]), in which vertices, instead of edges, are deleted. It was proven in [11] that the problem of finding optimal edge-irregulators is NP-hard even when restricted to planar bipartite graphs of maximum degree 6, and W[1]-hard when parameterized by the size of the solution, the feedback vertex set of the input graph, or its vertex integrity.

The first contribution of this work is the continuation of the study of the parameterized complexity of computing an optimal edge-irregulator of a given graph. Considering the previously mentioned W[1]-hardness results, we propose adding the maximum degree Δ of the input graph as part of the parameter. This proves to be fruitful, as we show that the problem becomes FPT w.r.t. the solution size plus Δ. We then present an FPT algorithm parameterized by the vertex cover number of the input graph, which is more efficient than the corresponding algorithm that is implied by the work in [11] (which is based on a reduction to an ILP with bounded number of variables).

The behaviour of computing I_e is less understood when it comes to structural parameters capturing dense graphs. In particular, it is unknown whether the problem is FPT when parameterized by parameters such as the neighbourhood diversity or the distance to clique of the input graph. We take an important step towards understanding the case of complete graphs. In particular, we prove that $\mathrm{I}_e(K_{t_k}) = |E(K_{t_k})| - \frac{k(k+1)(k-1)(3k+2)}{24}$, where t_k is any triangular number.

The study of this concept of edge-irregulators leads us to conjecture that there exists a constant c such that any connected graph G has an edge-irregulator of size at most $\frac{|E(G)|}{3} + c$. Note that the connected hypothesis is mandatory–the only edge-irregulator of a graph consisting only in a perfect matching will contain all the edges of the graph. Appart form attacking the case of complete graphs, we also prove this conjecture for various families of simple graphs, including trees, for which we also provide a dedicated polynomial algorithm.

2 Preliminaries, First Results and a Conjecture

We follow standard graph theory notations [9]. Let $G = (V, E)$ be a graph. For any vertex $v \in V$, let $N_G(v) = \{u \in V : uv \in E\}$ denote the *neighbourhood* of v in G, and let $d_G(v) = |N_G(v)|$ denote the *degree* of v in G. We also define

$N_G[v] = N_G(v) \cup \{v\}$. Whenever the graph G is clear from the context, we will omit the subscript and simply write $N(v)$, $d(v)$ and $N[v]$.

Parameterized complexity is a field in algorithm design that takes into consideration additional measures to determine the time complexity. The goal in this paradigm is to design a *Fixed-Parameter Tractable* (FPT) algorithm, i.e., an algorithm that solves the problem in $f(k)|x|^{O(1)}$ time for any arbitrary computable function $f\colon \mathbb{N} \to \mathbb{N}$, where $|x|$ is the size of the input and k is some bounded *parameter*. We say that a problem *is in* FPTif it admits an FPTalgorithm. A parameterized problem is *slicewise polynomial* if it can be determined in $|x|^{f(k)}$ time for a computable function $f\colon \mathbb{N} \to \mathbb{N}$. Such a problem then belongs to the class XP. A problem is presumably not in FPTif it is shown to be W[1]-hard (by a parameterized reduction). We refer the interested reader to classical monographs [8,10] for a more comprehensive introduction to this topic.

We are now ready to formally define the problem studied in this work. Let $G = (V, E)$ be a graph. We say that G is *locally irregular* if, for every edge $uv \in E$, it holds that $d_G(u) \neq d_G(v)$. Next, let $S \subseteq E$ and consider the graph G' that results from the deletion of the edges of S from G. We will denote this G' as $G - S$. Any set $S \subseteq E$ such that $G - S$ is locally irregular will be called called an *edge-irregulator of* G. Moreover, let $\mathrm{I}_e(G)$ be the minimum order that any edge-irregulator of G can have. We will say that S is an *optimal* edge-irregulator of G if S is an edge-irregulator of G and $|S| = \mathrm{I}_e(G)$. Our goal is to compute an optimal edge-irregulator of a given graph G (optimisation version) or, given an integer $k \geq 1$, decide if $\mathrm{I}_e(G) \leq k$ (decision version).

Irregularity-Deletion
Input: A graph G, and a positive integer $k \geq 1$.
Question: Do we have $\mathrm{I}_e(G) \leq k$?

To begin, observe that checking if a graph G is locally irregular can be done in polynomial time as it suffices to go through its edges once. Thus, checking if $\mathrm{I}_e(G) \leq k$ is (trivially) in XP when parameterized by k.

Intuitively, regular graphs should be the graph having the largest value of I_e. Asymptotically, if G is a cycle with m edges, then all its adjacent vertices are in conflicts and removing an edge resolves at most 3 of these conflicts. Thus, $\mathrm{I}_e(G)$ cannot be too far from $\frac{1}{3}m$. Equality cannot always hold due to divisibility matters; for instance, $\mathrm{I}_e(C_4) = 2$, which is $\frac{1}{2}m$ (where $m = |E(C_4)| = 4$) but also $\lfloor \frac{1}{3}m \rfloor + 1$. Even worse, $\mathrm{I}_e(K_2) = 1$, which is m (since $m = |E(K_2)| = 1$). Perhaps a good way to express our presumption is in the following form:

Conjecture 1. There is an absolute constant $c \geq 1$ such that, for every connected graph G with m edges, we have $\mathrm{I}_e(G) \leq \frac{1}{3}m + c$.

The additive constant c is mandatory to handle some uncontrolled behaviour of small and sparse graphs. In particular, our result on cycles (see Theorem 7) shows that $c \geq 2$ for any cycle C_{3k+2}, where k is an integer.



Fig. 1. Example of graph satisfying $\mathrm{I}_e(G) = \left\lceil \frac{\mathrm{conf}(G)}{2\Delta-1} \right\rceil$.

We want to insist on the fact that the connectivity requirement in Conjecture 1 is important, as the conjecture would be false otherwise. Consider, e.g., a disjoint union of several copies of K_2.

Throughout Sect. 4, we will prove that this conjecture holds for common classes of graphs. These points apart, Conjecture 1 is also supported by the fact that, when trying to construct connected graphs for which the parameter I_e is a "large" fraction of the number of edges, the best we could achieve is indeed a third of the number of edges. This also shows that Conjecture 1 is tight.

Theorem 1. *Let G be any (connected) graph on m edges obtained from a (connected) graph H by subdividing exactly twice every edge of H. Then, $\mathrm{I}_e(G) \geq \frac{1}{3}m$.*

As far as lower bounds are concerned, we observe that upon deleting an edge in a graph, the number of *conflicts*, *i.e.*, of remaining edges uv with $d(u) = d(v)$, can decrease by a limited amount only. Formally, for a graph G, we define $\mathrm{conf}(G)$ as the number of conflicts in G. We stress here the *static* nature of this definition: it only concerns G in its original state.

Remark 1. In any graph G with maximum degree $\Delta \geq 1$, deleting any edge decreases the number of conflicts by at most $2\Delta - 1$. Thus, $\mathrm{I}_e(G) \geq \left\lceil \frac{\mathrm{conf}(G)}{2\Delta-1} \right\rceil$.

The bound in Remark 1 is very general, and it can be bad for some graphs. As an illustration, consider $K_{n,n}$: the bound we obtain is $\mathrm{I}_e(K_{n,n}) \geq \frac{n^2}{2n-1}$. Later on, we will show this is distant from the optimal by about a factor of 2. There are other contexts, however, in which this bound is not too bad, as we will see later on with the case of paths and cycles. Nevertheless, the bound in Remark 1 can be tight for some graphs, as attested by the graph shown in Fig. 1.

3 FPT Algorithms

In this section we provide our efficient algorithms for computing an optimal edge-irregulator of a given graph.

3.1 Kernelization by Solution Size Plus the Maximum Degree

Theorem 2. Irregularity-Deletion *has a kernel of size $O(k\Delta^{2k+2})$ parameterized by the solution size k plus the maximum degree Δ of the input graph G, and thus an optimal edge-irregulator of G can be computed in FPT-time.*

The kernel that will be used in Theorem 2 will be mostly due to the following lemma, and the fact that we can bound the number of vertices in a graph in function of its diameter and its maximum degree.

Lemma 1. *Let $G = (V, E)$ be a graph, and let S be an optimal edge-irregulator of G. For any edge $e \in S$, there exists an edge $uv \in E$ such that $d(u) = d(v)$, and e is at distance at most $2|S| - 1$ from uv.*

Proof. The proof is by induction on $k = |S|$.

Assume first that $k = 1$. Since $S = \{e\}$ is an optimal edge-irregulator and is not empty, there exists at least one conflict $uv \in E$. Thus, since $G - \{e\}$ is locally irregular, either $e = uv$ or one vertex among u or v has a different degree in $G - S$, which implies that e is either adjacent to u or to v, and thus at distance at most 1 from e.

Suppose now $k \geq 2$. Let uv be a conflict in G. Since in $G - S$, the vertices u and v are not in conflict, there exists one edge e in S that is adjacent to u or to v. Consider now the graph $G - \{e\}$. By the optimality of S, we get that $S - \{e\}$ is an optimal edge-irregulator of $G - \{e\}$. Thus by induction hypothesis, all the edges of S are at distance at most $2|S - \{e\}| - 1 = 2|S| - 3$ from a conflict in $G - \{e\}$. Consider now a conflict ab in $G - \{e\}$. Then either ab was already a conflict in G, or ab has become a conflict due to the deletion of e. This means that either a or b is adjacent to wither u or to v, and thus ab is at distance at most 2 from uv. Finally, using the triangular inequality, any edge of S is at distance at most $2k - 3 + 2 = 2k - 1$ from a conflict of G.

3.2 Vertex Cover Number

While it is already known that $I_e(G)$ can be computed in FPT-time parameterized by the vertex cover number using the FPT-algorithm for the more general parameter of *vertex integrity* [11], we present here an algorithm that is tailor-made for the vertex cover number, and, as such, is faster.

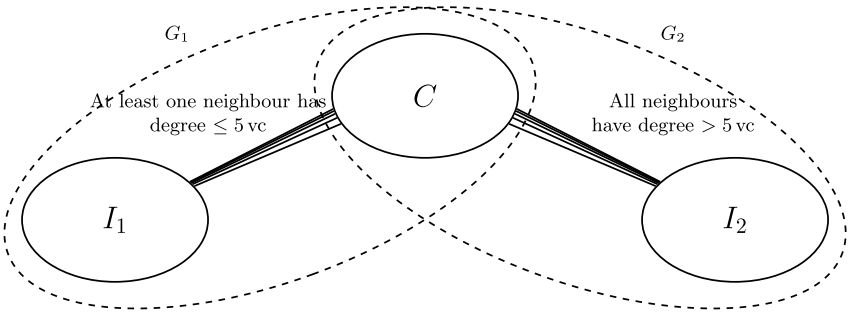


Fig. 2. An abstract depiction of the graph G of Theorem 3.

Theorem 3. *There exists an algorithm to compute $I_e(G)$ in time $2^{O(vc^4)} \cdot n^{O(1)}$, where vc is the size of a minimum vertex cover of G and n is the order of G.*

Proof. We first describe the algorithm and then prove its correctness and time-complexity.

Let $G = (V, E)$ be the given graph. Also, let $C = \{u_1, \ldots, u_{vc}\}$ be a minimum vertex cover of G and $I = V \setminus C$ be an independent set. We partition I into I_1 and I_2 as follows. For every vertex $u \in C$, if $d_I(u) \leq 5vc$, we include all the vertices of $N_I(u)$ into I_1. Once we have finished considering all the vertices of C, we place the remaining vertices of I into I_2. Let $G_1 = G[C \cup I_1]$ and $G_2 = G[C \cup I_2]$ (see Fig. 2). We will show later that this results in $|I_1| \leq 5vc^2$ and $|V(G_1)| \leq 5vc^2 + vc$. We can now guess the projection S_1 of an optimal edge-irregulator S of G onto G_1. That is, $S_1 = S \cap E(G_1)$. Notice that $G_1 - S_1$ is not necessarily locally irregular. All that remains to do is to compute the smallest set $S_2 \subseteq E(G_2)$ such that $G - (S_1 \cup S_2)$ is locally irregular. Notice that such a set S_2 contains only edges between the vertices of C and I_2; let E_2 denote these edges.

Henceforth, we focus on G_2. For every $i \in [vc] = \{1, \ldots, vc\}$, we define a *modifier* $k_i \in [n]$. These modifiers encode the number of edges of E_2 that need to be removed in order to arrive to a locally irregular graph. That is, if k_i edges of E_2 incident to u_i are removed from G_2, then u_i is not in conflict with any of its neighbours (in G). Note that for each i, the modifier k_i does not necessarily have a unique value. We will nevertheless show that it is sufficient to compute a minimum (with respect to summation) tuple $(k_1, \ldots, k_{vc})$ such that $\Sigma_{i=1}^{vc} k_i \leq vc^2$.

Correctness. We first argue about the size of I_1.

Claim 4. *We have that $|I_1| \leq 5vc^2$.*

Proof of the Claim. Consider how I_1 is constructed. By definition of I_1, every vertex of C adjacent to a vertex of I_1 has degree at most $5vc$. Thus, $|I_1| = |\{N[v] | v \in C, d(v) \leq 5vc\}| \leq 5vc \cdot |C| = 5vc^2$.

Claim 5. *Let $S_1 \subset E_1$ be a set of edges. For any optimal edge-irregulator S of G coinciding with S_1 on E_1, we have $S \cap E_2 \leq vc^2$.*

Proof of the Claim. Let S be an optimal edge-irregulator of G coinciding with S_1 on E_1. Assume by contradiction that $S \cap E_2 > vc^2$. Since all the edges of E_2 has one endpoint in C, by the pigeonhole principle, there exists a vertex v in C adjacent to at least $vc+1$ edges of E_2. If at least $2vc+1$ edges of S are adjacent to v, then we first remove vc of them from S. In this case, S may not yet be an edge-irregulator but the only conflicts we may have created are between v and other vertices of C (since all the other neighbours of v are not in C and thus have degree at most vc). Either way, since the other vertices of C can forbid at most $vc - 1$ conflicts, there exist a number $1 \leq k \leq vc$ such that increasing the degree of v by k removes all these conflicts. Since there are at least $vc+1$ vertices of $S \cap E_2$ that are still adjacent to v, we can remove k of them to solve all the

conflicts. This leads to a smaller edge-irregulator S' and $S' \cap E_1 = S \cap E_1 = S_1$, which is a contradiction to the optimality of S.

Claim 6. *Let S be an optimal edge-irregulator of G, and u, v_1, v_2 be three vertices with $u \in C$ and $v_1, v_2 \in I_2$. If $uv_1 \in S$, $uv_2 \in E$ and $uv_2 \notin S$, then there exists an optimal edge-irregulator S' of G such that $S' = (S \setminus \{uv_1\}) \cup \{uv_2\}$.*

Proof of the Claim. First, note that the proof of Claim 5 also proves that if S is an optimal edge-irregulator of G, for any vertex $u \in C$ which has neighbours in I_2, we have $d_{G-S}(u) > vc$.

Consider now $S' = (S \setminus \{uv_1\}) \cup \{uv_2\}$ and let $G' = G - S'$. Since $v_1, v_2 \in I_2$, their degrees are at most vc and all their neighbours have a degree of at least $vc + 1$. Thus S' does not create any conflicts for v_1 or for v_2. Moreover, all the other vertices in G have the same degree and the same neighbourhood in $G - S$ and in $G - S'$. Thus, since S was an edge-irregulator, they are not in conflict with each other. Thus, S' is an edge-irregulator.

Running Time. By Claim 4, we have that $|C \cup I_1| \leq vc + 5vc^2$. Thus, there are at most $2^{(vc+5vc^2)^2}$ possible guesses of S_1. By Claim 5, we know that an optimal edge-irregulator of G has at most vc^2 edges in E_2, hence $|S_2| \leq vc^2$. Due to Claim 6, it suffices to guess for each vertex $u \in C$ how many edges of E_2 incident to u are in S_2, that is, a tuple of integers $(k_1, \ldots, k_{vc})$ such that $k_1 + \cdots + k_{vc} \leq vc^2$. There are vc^{vc^2} such possibilities. Recall that checking whether a set $S \subset E(G)$ is an edge-irregulator can be done in running time n^2 where n is the order of G, and since we repeat this step for each guess of S_1, the running time of this algorithm is $2^{(vc+5vc^2)^2} \cdot vc^{vc^2} \cdot n^2 = 2^{O(vc^4)} \cdot n^{O(1)}$. □

4 Structural Results

In this section we investigate common classes of graphs to determine their optimal edge-irregulators. Our investigation is lead by Conjecture 1, which we verify for most of the considered classes. The results in this section are important in order to gain a better understanding of I_e in regards to structural parameters, but also since general bounds on I_e can be deduced from simpler cases:

Remark 2. Let G be a graph and S be a set of edges of G such that $G-S$ consists of $p \geq 1$ connected components $G_1, \ldots, G_p$. Then, $\mathrm{I}_e(G) \leq |S| + \sum_{i=1}^{p} \mathrm{I}_e(G_i)$.

We start by considering graphs with maximum degree 2 (paths and cycles). For $n \geq 2$, we denote by P_n the path of order n, and, for $n \geq 3$, by C_n the cycle of order n. In this case, we are able to determine I_e completely. Roughly, to prove this result we delete the minimum number of edges to reach as many disjoint P_3s as possible.

Theorem 7. *For every $n \geq 2$, we have:*

- $\mathrm{I}_e(P_n) = \frac{n-1}{3}$ *if* $n \equiv 1 \bmod 3$;
- $\mathrm{I}_e(P_n) = \lceil \frac{n-1}{3} \rceil$ *if* $n \equiv 2 \bmod 3$;

- $\mathrm{I}_e(P_n) = \lfloor \frac{n-1}{3} \rfloor$ *otherwise.*

Moreover, for every $n \geq 3$*, we have* $\mathrm{I}_e(C_n) = \mathrm{I}_e(P_n) + 1$.

Next, we deal with complete bipartite graphs. Recall that $K_{n,m}$ denotes the complete bipartite graph with parts of size n and m ($n, m \geq 1$).

Theorem 8. *For every* $n, m \geq 1$*, we have:*

- $\mathrm{I}_e(K_{n,m}) = 0$ *if* $n \neq m$*;*
- $\mathrm{I}_e(K_{n,m}) = n$ *otherwise.*

Proof. The first item is because $K_{n,m}$ is locally irregular whenever $n \neq m$. We now focus on $K_{n,n}$ with $n \geq 2$. Choose any vertex, u, of $K_{n,n}$, and let S be the set containing all edges incident to u. Then, clearly, $K_{n,n} - S$ is locally irregular, and $\mathrm{I}_e(K_{n,n}) \leq n$. To see, now, that $\mathrm{I}_e(K_{n,n}) \geq n$, consider, towards a contradiction, any set S of at most $n-1$ edges such that $K_{n,n} - S$ is locally irregular. Denoting by (U, V) the bipartition of $K_{n,n}$ (where, thus, $|U| = |V| = n$), since $|S| \leq n-1$, there must be at least a $u \in U$ not incident to any edge of S, and similarly a $v \in V$ not incident to any edge of S. Then, $d_{K_{n,n}-S}(u) = d_{K_{n,n}-S}(v) = n$, while u and v are adjacent in $K_{n,n} - S$; this is a contradiction. □

We now move towards more general families of graphs.

Theorem 9. *If* $G = (V, E)$ *is a connected bipartite graph of order* n *and minimum degree* 1*, then* $\mathrm{I}_e(G) \leq n - 1$.

Sketch. Let r be any degree-1 vertex of G, and let $\mathcal{V} = \{V_1, V_2, \dots V_d\}$ be the partition of $V(G) \setminus \{r\}$ where each V_i contains the vertices of G at distance exactly i from r. We construct a set $S \subseteq E$ such that, in $G - S$, every vertex in V_i has a degree of the same parity as i (except maybe for vertices becoming isolated, and one vertex; see later). We start with S being empty. We consider the vertices of $V(G) \setminus \{r\}$ one by one, following the parts $V_d, V_{d-1}, \dots, V_1$ in this order. For any vertex $u \neq r$ considered this way, if $d_{G-S}(u)$ does not have the required parity, we add to S a single edge uu' where u' is closer to r than u is. Once all vertices of $\mathcal{V}$ have been treated this way, the only potential issue can come from r having the same degree as its sole neighbour w in $G - S$, which is dealt with easily since r is a degree-1 vertex. ◇

Adapting the above strategy, we can prove that:

Theorem 10. *For every connected graph* G *of order* n*, with* m *edges and maximum degree* Δ*, we have* $\mathrm{I}_e(G) \leq \lfloor \frac{m}{2} \rfloor + n + \Delta - 2$.

4.1 Trees

We now focus on trees. We first prove that Conjecture 1 holds for this family, and then provide a linear-time algorithm for trees of bounded maximum degree.

We proceed in two steps: we first deal with "star-like" trees, before treating the more general case. Recall that a *bistar* is a tree with exactly two vertices u

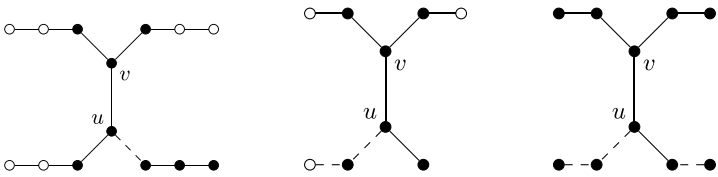


Fig. 3. A representation of the case where $d(u) = d(v) = 3$ and u, v are adjacent in the proof of Lemma 2. A white vertex may or may not exist. The dashed edges form an optimal edge-irregulator. In the leftmost figure, we do not provide an edge-irregulator, but rather fall back in the case of a subdivided star.

and v of degree at least 3 that are adjacent. See Fig. 3 for a representation of a bistar, as well as some intuition on the proof of the following lemma. For the more general case, we also need to observe a lemma concerning bridges.

Lemma 2. *If G is a subdivision of a star or of a bistar, then $\mathrm{I}_e(G) \leq \frac{1}{3}|E(G)|$.*

Lemma 3. *Let S and S' be two edge-irregulators of two graphs G, G', respectively, and let $u \in V(G)$ and $v \in V(G')$. Let H be the graph obtained from the disjoint union of G and G' with the addition of the edge uv. Then $I = S \cup S' \cup \{uv\}$ is an edge-irregulator of H.*

Theorem 11. *Let T be a tree. Then either T is a path or $\mathrm{I}_e(T) \leq \frac{1}{3}|E(T)|$.*

Proof. Suppose by contradiction that G is a minimum counterexample to this theorem. Then G cannot be a path, and by Lemma 2, G cannot be a subdivision of a star or of a bistar. We root G at an arbitrary vertex r, and we consider u a deepest vertex of degree $k \geq 3$ in G. If several exist, we choose one of the highest degree. Denote by v its parent. Note that $u \neq r$, otherwise the graph would be a subdivided star. If $v = r$, we chose a neighbour of r other than u as the new root. For $w \in V(G)$, we will denote by T_w the subtree induced by w and all its descendant for this rooting.

Claim 12. *Let $P = (p_1, p_2, p_3)$ be a path of G with $d(p_1) = 1$ and $d(p_2) = 2$. Then $d(p_3) \geq 3$.*

Proof of the Claim. Since G is connected, $d(p_3) \neq 1$, otherwise we would have $T = P$, which contradicts that G is not a path. Suppose by contradiction $d(p_3) = 2$, let $G' = G \backslash T_{p_3}$ and let S' be an edge-irregulator of G'. By minimality of G, we have that $|S'| \leq \frac{|G'|}{3}$. Let p'_3 be the parent of p_3, and consider $S = S' \cup \{p_3 p'_3\}$. Then S is an edge-irregulator of G of size at most $\frac{|G|}{3}$, a contradiction.

It follows that, by construction, the subtree T_u rooted in u is a subdivided star. Let $P_i = (v_{i,1}, \ldots, v_{i,k_i})$ for $i \in [k-1]$ be its branches. Without loss of generality, and according to Claim 12, we can suppose for $1 \leq i < j \leq k$, $|P_i| \geq |P_j|$, and thus $2 \geq |P_1| \geq \cdots \geq |P_k| \geq 1$.

- If $k \geq 4$, then by the minimality of G, we get that $G' = G \setminus T_u \cup \{u\}$ has an edge-irregulator S' of size at most $\frac{1}{3}|E(G')|$. If $uv \in S'$, then $S \cup S'$ is an edge-irregulator of the required size, where S is an edge-irregulator of T_u of size at most $\frac{|T_u|}{3}$ obtained from Lemma 2. Now suppose $uv \notin S'$. In $G \setminus S'$, the only remaining conflicts can be the edge uv. Should this happen, all the edges from P_{k-1} form an edge-irregulator of T_u decreasing the degree of u by one, thus resolving the conflict uv, and $|P_i| \leq \frac{|T_u|}{3}$.
- We can now assume $k = 3$. There are thus two induced paths P_1, P_2 rooted at u. We will distinguish cases depending on the lengths of $|P_1|$ and $|P_2|$.
  - Assume $|P_1| = |P_2| = 1$ and let $G' = G \setminus T_u$. By minimality of G, the tree G' is not a path and has an edge-irregulator $S'$of size $\frac{|G'|}{3}$. By construction, $S = S' \cup \{uv\}$ is an edge-irregulator of G. Moreover, we have $|S| = |S'| + 1$ and $|G| = |G'| + 3$.
  - Assume now $|P_1| = 2$ and $|P_2| \in \{1, 2\}$. Let v' be the parent of v. If $d_G(v) = 1$, then the graph is already locally irregular and we are done. If $d_G(v) = 2$ and G' is a path then we are done by Lemma 2. Otherwise (if $d_G(v) = 2$ and G' is not a path), we consider $G' = G - T_v$ and let S' be an optimal edge-irregulator of G'. Using Lemma 3, the set $S = S' \cup \{vv'\}$ is an edge-irregulator of G and has size less than $\frac{1}{3}|E(G)|$.
    We can now assume $d(v) \geq 3$. By the definition of u, any other descendant of v of degree at least 3 must be a neighbour of v. Moreover, by minimality of G, we can suppose that all its other neighbours of degree 3 cannot fall into the previous cases. Thus, we can suppose that all its other neighbour have degree at most 3, and if they have degree 3, they are adjacent to a path of length two and a path of length at most 2. Moreover, we recall that by minimality of G, there cannot be a path of length at least 3 attached to v.
    - Assume first $d_G(v) \geq 5$ and let $G' = G - T_v$. Consider S' an optimal edge-irregulator of G'. Note that G' can be a path, but a path has an edge-irregulator of size at most $\frac{|G'|}{3} + 1$ which will be sufficient here. Let $S = S' \cup \{vv'\}$. It is easy to check that $G - S$ is indeed locally irregular. Since exactly one edge adjacent to v was removed, its resulting degree is at least 4, all its neighbours have degree at most 3 and the rest of T_v was already locally irregular. Moreover, T_v contains at least 7 edges. Thus $|S| = |S'| + 1 \leq \frac{|G'|}{3} + 2 \leq \frac{|G|-7}{3} + 2 \leq \frac{|G|}{3}$
    - Assume $d(v) = 3$ and let w be the neighbour of v other than u and v'. We consider $G' = G - T_v$. If G' is a path, then $d_g(w) = 3$, as otherwise G would be a bistar. In this case, consider S' an edge-irregulator of G' of size at most $\frac{|G'|}{3} + 1$ and let $S = S' \cup \{vv'\}$. As in the previous case, since there are at least 8 edges in T_v and since T_v is already locally irregular, we are done. Otherwise, G' is not a path and we can consider S' an optimal edge-irregulator of G' of size at most $\frac{|G'|}{3}$. If $d_G(w) \in \{1, 3\}$, then $S = S' \cup \{vv'\}$ is an edge-irregulator of size at most $\frac{|G|}{3}$. Otherwise, $d_G(w) = 2$. Denote by w' the child of w in T_v. Note that w' is a leaf, since we have already proved that there is no

path of length at least 3 adjacent to v. Thus, $S \cup \{vv', ww'\}$ is an edge-irregulator of G of size at most $\frac{|G|}{3}$ as $|T_v| \geq 6$.

- We can finally assume $d(v) = 4$. Let w and x be the two other children of v. W.l.o.g., suppose $d(w) \geq d(x)$. By hypothesis on u, we have $3 = d(u) \geq d(w) \geq d(x)$, and both w and x can only have paths as descendents. In case of equality in their degrees, suppose $|T_u| \geq |T_w| \geq |T_x|$.
  - If $d(w) = 3$, let $G' = G \setminus T_v$. If G' is a path, G is already locally irregular. Indeed, all the vertices of degree 3 or more are v and some of its neighbours. Thus, using Claim 12, and since all the neighbours or v have degree at most 3, there are no conflicts in G. Otherwise, let S' be an edge-irregulator of G', and S_x be an optimal edge-irregulator of T_x. Since G' is not a path and G is a minimum counter-example, we have $|S'| \leq \frac{|G'|}{3}$. Also, $|S_x| \leq \left\lceil \frac{|T_x|}{3} \right\rceil$ since T_x is a path. Consider $S = S' \cup \{vv', vx\} \cup S_x$. By construction, S is an edge-irregulator of G, and $|S| \leq \frac{|G|}{3}$, see Fig. 4(a).
  - If $d_G(w) = 2$ and $d_G(x) = 2$, let $G' = G \setminus (T_u \cup T_w \cup T_x)$ and let S' be an optimal edge-irregulator of G'. If $d_{G'-S'}(v') \neq 4$, since $d_{G'-S'}(v) = 4$, we have that S' is also an edge-irregulator of G and has size at most $\frac{G}{3}$. Otherwise, let S_u be an optimal edge-irregulator of T_u. Note that T_u is either a path on 4 or on 5 vertices, which implies $|S_u| = 1$ or $|S_u| = 2$, respectively. Consider $S = S' \cup S_u \cup \{(vu)\}$. Then S is an edge-regulator of G, and $|S| = |S'| + 1 + |S_u| \leq \frac{|G'|}{3} + 1 + |S_u| \leq \frac{|G|}{3}$ since $|T_v| = 8$ if $|T_u| = 4$ and $|T_v| = 9$ if $|T_u| = 5$, see Fig. 4(b).
  - Finally, suppose $d_G(w) \leq 2$, and $d_G(x) = 1$. Let $G' = G \setminus T_v$. Once again, if G' is a path, G is already locally irregular. Let S' be an edge-irregulator of G', and let $S = S' \cup \{vv', vx\}$. If $d(w) = 2$, we also add to S the edge connecting w to its leaf. In both cases, S is an edge-irregulator of G of size at most $\frac{G}{3}$, see Fig. 4(c).

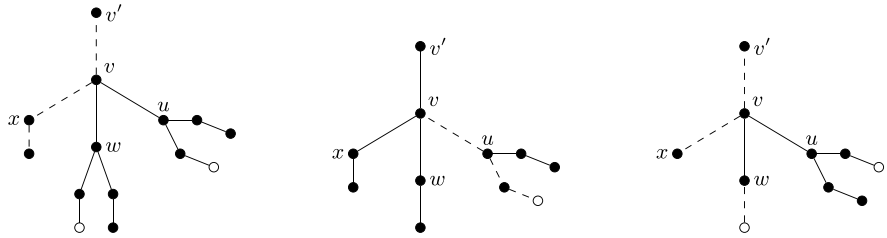
Thus, a minimal counterexample does not have a vertex of degree at least 3, meaning it is a path, which concludes the proof.

Theorem 11 proves that Conjecture 1 holds for trees. For this family we can also compute the exact value of $\mathrm{I}_e(G)$ with an efficient algorithm.

Theorem 13. *If G is a tree of order n and maximum degree Δ, then computing $\mathrm{I}_e(G)$ can be done in time $O(n\Delta^3)$.*

4.2 Complete Graphs

For any $n \geq 3$, we denote by K_n the complete graph of order n. Complete graphs are an interested case for our problem, as computing $\mathrm{I}_e(K_n)$ is the same as finding the size of locally irregular graphs of order n with the most edges.



(a) $d(w) = 3$, $d(x) = 2$, $|P_1| = 2$ and $|P_2| \leq 2$ (b) $d(w) = 2$, $|P_1| = 2$, $d(x) = 2$ and $|P_2| \leq 2$ (c) $d(w) = d(x) = 2$ and $|P_1| = |P_2| = 2$

Fig. 4. A representation of some cases of the proof of Theorem 11. The white vertices may or may not exist. The dashed edges are the edges in S.

In what follows, we are solving the question for particular values of n, while we only provide partial results for the remaining ones. Precisely, we focus on the case where n is a *triangular number*, *i.e.*, of the form $\frac{k(k+1)}{2}$ for some k. For any $k \geq 1$, we denote by t_k the kth triangular number, being the sum of all integers from 1 up to k.

Lemma 4. *If $n = t_k$ is a triangular number, then the maximum number of edges in a locally irregular graph of order n is exactly $m_k := \frac{k(k+1)(k-1)(3k+2)}{24}$.*

Sketch. Let G be a locally irregular graph of order $n = t_k$. Observe that if G is a locally irregular graph of order n, then G has at most k vertices of degree $n - k$ for all $k \in [k]$. Thus, for every $i \in [k]$, there are at most i vertices of degree $n - i$ in G. So, any locally irregular graph of order n having the most edges has at most one vertex of degree $n - 1$, two vertices of degree $n - 2$, and so on. Because n is a triangular number, such a graph T_k can be constructed for all $k \geq 1$; we prove this by induction on k. The graph T_1 is the graph containing a single vertex. Then, for any $i \geq 1$ such that T_i has been constructed, we obtain T_{i+1} from T_i by adding $i + 1$ independent vertices adjacent to all the vertices of T_i. It is easy to see that for any $k \geq 1$, the graph T_k indeed has the claimed degree sequence. To fully prove the claim, it remains to express the number of edges of any T_k, which is achieved through the handshaking lemma. ◇

Theorem 14. *For every $k \geq 1$ and $n = t_k$, we have $\mathrm{I}_e(K_n) = |E(K_n)| - m_k$.*

Even though Theorem 14 is tight, it cannot be generalised easily to complete graphs whose order is not a triangular number. However, since $t_{k+1} - t_k = k = \Theta(\sqrt{t_k})$ for every $k \geq 1$, Lemma 4 yields both a lower and an upper bound on the size of locally irregular graphs of order n having the most edges, when n is not triangular: namely, they have $\frac{n^2}{2} - O(n^{3/2})$ edges. And from this, we proved that Conjecture 1 holds for all complete graphs.

5 Conclusion

In this work, we extended the study of edge-irregulators from [11]. Apart from focusing on the algorithmic aspects of this problem, we also derived general bounds and analysed behaviours of optimal edge-irregulators, leading to Conjecture 1, which we verified for several graph classes.

Determining $\mathrm{I}_e(G)$ for a graph G remains challenging due to the intricate properties and elusive structure of maximally edge-dense locally irregular graphs. Let us denote by $\mathcal{L}_n$ (for some $n \geq 3$) the class of all locally irregular graphs of order n with the most edges possible. In light of Lemma 4, a key open question concerns the size and structural properties of $\mathcal{L}_n$ in the case where n is a non-triangular number.

We finally observe that there is room for improvement for many of our results. This could lead to further work on the topic. In particular, we find the following questions appealing.

- Refining the static parameter conf (Sect. 4) to dynamically capture evolving edge conflicts.
- Determining I_e for complete graphs and tightening bounds for cubic graphs.
- Extending Theorem 1 to subcubic graphs. Also, for k-degenerate graphs G of order n, we could get an upper bound on $\mathrm{I}_e(G)$ of about $(k-1)n + \left\lfloor \frac{n}{3} \right\rfloor$ through exploiting similar ideas to the ones used in this paper.

It is also interesting to further study the algorithmic complexity of determining $\mathrm{I}_e(G)$ for a given graph G. In particular, the authors of [12] proved that the vertex counterpart of the problem is in FPT when parameterized by the treewidth and the maximum degree. Their algorithm should adapt in an obvious way to our context. Thus, we prefer to not include such a result here.

We finish by commenting on the FPT algorithm we present here w.r.t. the vertex cover number. First, observe that there is still room for improvement in its running time. Indeed, we are only aware of a $2^{o(vc)}n^{O(1)}$ lower bound (based on the ETH) for this problem (stemming as a direct corollary of the W[1]-hardness w.r.t. the solution size proof presented in [11]). Lastly, we believe that this algorithm could be generalised to work w.r.t. the neighbourhood diversity of the input graph. However, in order to achieve this, it would be mandatory to completely determine I_e for complete graphs.

References

1. Alavi, Y., Chartrand, G., Chung, F., Erdős, P., Graham, R., Oellermann, O.: Highly irregular graph. J. Graph Theory **11**(2), 235–249 (1987)
2. Baudon, O., Bensmail, J., Przybyło, J., Woźniak, M.: On decomposing regular graphs into locally irregular subgraphs. European J. Combin. **49**, 90–104 (2015)
3. Bazgan, C., Cazals, P., Chlebíková, J.: Degree-anonymization using edge rotations. Theoretical Comput. Sci. **873**, 1–15 (2021)
4. Belmonte, R., Sau, I.: On the complexity of finding large odd induced subgraphs and odd colorings. Algorithmica **83**(8), 2351–2373 (2021)

5. Bensmail, J., Li, B., Li, B.: An injective version of the 1-2-3 conjecture. Graphs Combin. **37**, 281–311 (2021)
6. Bensmail, J., Catherinot, N., Fioravantes, F., Marcille, C., Oijid, N.: Graph Irregularity via Edge Deletions (2025). https://hal.science/hal-05084434
7. Chartrand, G., Jacobson, M., Lehel, J., Oellermann, O., Ruiz, S., Saba, F.: Irregular networks. Congr. Numer. **64**, 197–210 (1988)
8. Cygan, M., et al.: Parameterized Algorithms. Springer, Cham (2015). https://doi.org/10.1007/978-3-319-21275-3
9. Diestel, R.: Graph Theory. GTM, vol. 173. Springer, Heidelberg (2017). https://doi.org/10.1007/978-3-662-53622-3
10. Downey, R.G., Fellows, M.R.: Fundamentals of Parameterized Complexity. TCS, Springer, London (2013). https://doi.org/10.1007/978-1-4471-5559-1
11. Fioravantes, F., Melissinos, N., Triommatis, T.: Parameterised distance to local irregularity. In: 19th International Symposium on Parameterized and Exact Computation (IPEC 2024). Leibniz International Proceedings in Informatics (LIPIcs), vol. 321, pp. 18:1–18:15. Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2024)
12. Fioravantes, F., Melissinos, N., Triommatis, T.: Complexity of finding maximum locally irregular induced subgraphs. Discret. Appl. Math. **363**, 168–189 (2025)
13. Karoński, M., Łuczak, T., Thomason, A.: Edge weights and vertex colours. J. Combin. Theory Seri. B **91**, 151–157 (2004)
14. Keusch, R.: A solution to the 1-2-3 conjecture. J. Comb. Theory Seri. B **166**, 183–202 (2024)